INCH-POUND

MIL-PRF-27/313C
W/AMENDMENT 1
31 March 2021
SUPERSEDING
MIL-PRF-27/313C
25 July 2008

PERFORMANCE SPECIFICATION SHEET

# TRANSFORMERS, POWER, 10 VOLT-AMPERES, 60 HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-PRF-27.

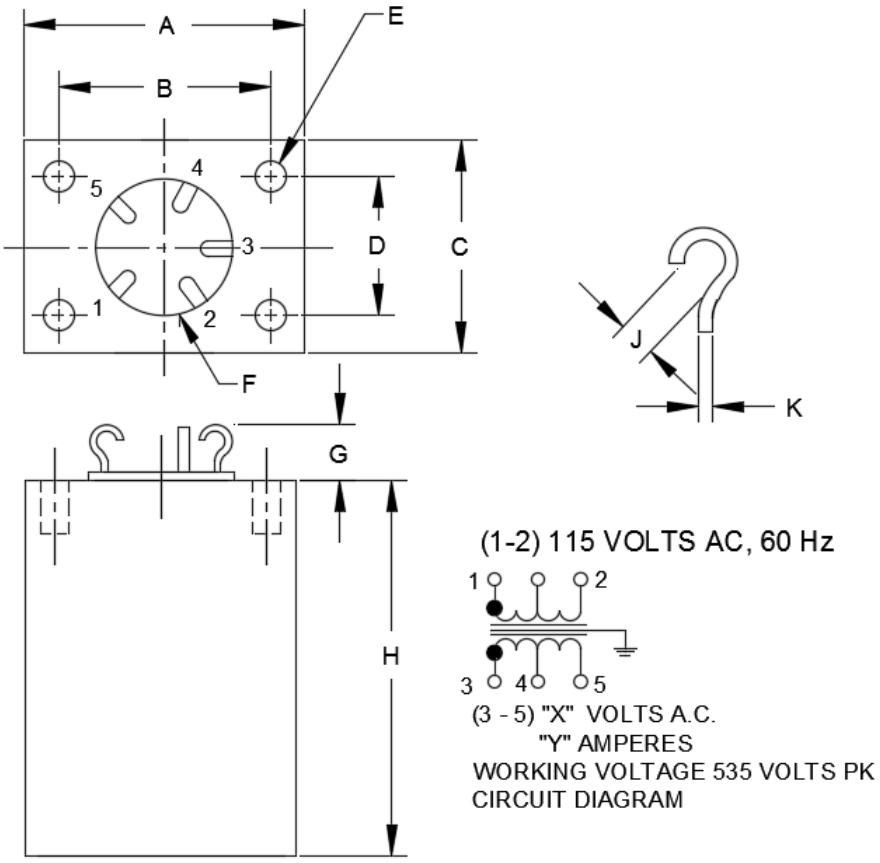



FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5950

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified the tolerance shall be ±.063 inch (1.60 mm).
5. Electrical values shall be marked as specified in table I, except DCR.

| Dim | Inches | mm |
|---|---|---|
| A | 2.06 ±.031 | 52.3 ±0.79 |
| B | 1.50 ±.016 | 38.1 ±0.41 |
| C | 1.75 ±.031 | 44.5 ±0.79 |
| D | 1.25 ±.016 | 31.75a ±0.41 |
| E | .138-32 UNC-2B (4 PLACES)<br>8 FULL THDS MIN | |
| F | .875 DIA | 22.23 DIA |
| G | .375 | 9.53 |
| H | 2.75 | 69.9 |
| J | .070 ±.005 | 1.78 ±0.13 |
| K | .062 ±.005 | 1.57 ±0.13 |

FIGURE 1. Dimensions and configurations – Continued.



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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts ac, 50 to 60 hertz.

Primary dc resistance: 36 ohms ±25 percent.

Secondary voltage (3-5): See table I.

Volt-ampere rating: 10 volt-amperes.

Working voltage (1-2): 535 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Terminals: Solderable hook terminals.

Terminal height: 0.375 ±0.063 inch.

Terminal diameter: 0.062 ±0.005 inch.

Weight: 1.5 pounds, maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 10,000 feet.

Terminal strength: MIL-STD-202-211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding) (at sea level): 1,500 volts rms.

Electrical characteristics:

Rated load: With 115 volts ac and 60 Hz across (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I.

Regulation: $\frac{\text{Voltage (no load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

Shall not exceed 12 percent.

Polarity: Additive with terminals 2 and 3 connected.



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Temperature rise: 35°C with 115 volts ac, 60 hertz across (1-2) at an ambient temperature of 95°C, full load terminals (3-5).

Marking location: See figure 1.

Part or Identification Number (PIN): M27/313-(dash number from table I).

TABLE I. Electrical ratings.

| Dash number | Secondary (3-5) | | |
|---|---|---|---|
| | Voltage (volts) ±3 percent | Current 1/ (amperes) | DC resistance (ohms) ±25 percent |
| 01 | 5.0 | 2.00 | 0.12 |
| 02 | 6.3 | 1.58 | 0.18 |
| 03 | 8.0 | 1.25 | 0.27 |
| 04 | 10.0 | 1.00 | 0.42 |
| 05 | 12.6 | 0.79 | 0.65 |
| 06 | 14.0 | 0.72 | 0.74 |
| 07 | 16.0 | 0.68 | 1.08 |
| 08 | 18.0 | 0.55 | 1.42 |
| 09 | 20.0 | 0.50 | 1.54 |
| 10 | 22.0 | 0.45 | 1.90 |
| 11 | 25.0 | 0.40 | 2.55 |
| 12 | 28.0 | 0.36 | 3.00 |
| 13 | 31.0 | 0.32 | 3.80 |
| 14 | 35.0 | 0.29 | 4.50 |
| 15 | 39.0 | 0.26 | 6.30 |
| 16 | 43.0 | 0.23 | 7.00 |
| 17 | 48.0 | 0.21 | 9.30 |
| 18 | 55.0 | 0.18 | 11.0 |
| 19 | 63.0 | 0.16 | 16.4 |
| 20 | 70.0 | 0.143 | 19.0 |
| 21 | 80.0 | 0.125 | 25.0 |
| 22 | 90.0 | 0.111 | 30.4 |
| 23 | 100 | 0.100 | 40.5 |
| 24 | 110 | 0.091 | 44.5 |
| 25 | 123 | 0.081 | 70.0 |
| 26 | 138 | 0.073 | 80.0 |
| 27 | 157 | 0.064 | 100 |
| 28 | 175 | 0.057 | 111 |
| 29 | 200 | 0.050 | 160 |
| 30 | 220 | 0.045 | 168 |

1/ For 50 Hz operation, reduce secondary current by 10 percent.



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VERIFICATION:

Extent of Qualification:

Qualification testing and approval to M27/312-26 or M27/316-30 or M27/322-24 shall be sufficient to grant qualification approval to MIL-PRF-27/312 through MIL-PRF-27/322, inclusive, all parts.

Qualification testing and approval to M27/313-30 shall be sufficient to grant qualification approval M27/313-01 through M27/313-30.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-211

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2021-010)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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